ACE Cyber Testbed Setup & Configuration

Purpose:
This document provides a structured record of how the research testbed environment was designed, configured, and validated to support your project's experiments. It ensures your environment is **reproducible**, **secure**, and **aligned with research goals**.

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1. Project Context

Field	Response
Student Name(s)	Denzel Jefferson
Research Title	Contextualized Threat Intelligence Prioritization for CTI Analysts
Faculty Advisor	Professor Julie Henderson
DoD/ACE Mentor	Professor William Littleton
Version	v3.0
Date	May 4, 2026

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2. Testbed Purpose and Scope

Describe the **overall purpose** of the testbed. Explain how it supports your research question and hypothesis.

Include:

- What is the **goal** of this testbed? (e.g., simulate SIEM alert handling, test anomaly detection, etc.) The goal of this testbed is to use a simulated CTI analyst environment that incorporates simulated operational environmental data into Elasticsearch and a simulated contextual scoring model. This environment supports the research question by evaluating an automated, contextualized threat-prioritization workflow against manual and traditional approaches.
- What research **variables** or conditions will be studied here? The research variables studied in this experiment include the use of contextual operational data, alert fatigue or volume reduction, the efficiency of CTI analyst workflows, and the accuracy of threat or alert prioritization ranking. These experiments will analyze the differences between basic threat rankings and automated, contextualized rankings.
- What **outcomes** do you expect from experiments in this environment? The expected outcomes of experiments conducted in this environment include reduced alert

fatigue, increased CTI analyst efficiency, and improved automated prioritization of high-impact threats.

- What **limitations** does this environment intentionally impose (e.g., isolated network, simulated data)? This testbed will intentionally impose limitations to comply with protection guidelines. The environment will be fully simulated, with isolated networks, synthetic logs, and no real users or production data. These factors will ensure that the experiments can be conducted in accordance with university guidelines and DoD data protection policies.

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3. Testbed Architecture Overview

Provide a **high-level diagram** or description of your testbed topology. The testbed architecture will be a virtual environment hosted on a personal workstation. The environment will consist of virtual machines and containers that will imitate the CTI analyst workflow, including log generation, analysis, and visualization. VirtualBox will be used to host Linux virtual machines, while Docker containers will be used to simulate log generation. All tools used will be hosted on a private network to ensure compliance with ethical research and university policies.

Include:

- All **nodes, VMs, or containers** and their roles
- Network segmentation and connectivity (firewalls, VLANs, bridges)
- Any **virtualization platforms** (VMware, VirtualBox, KVM, etc.)
- Any **container orchestration tools** (Docker, Podman, Kubernetes)

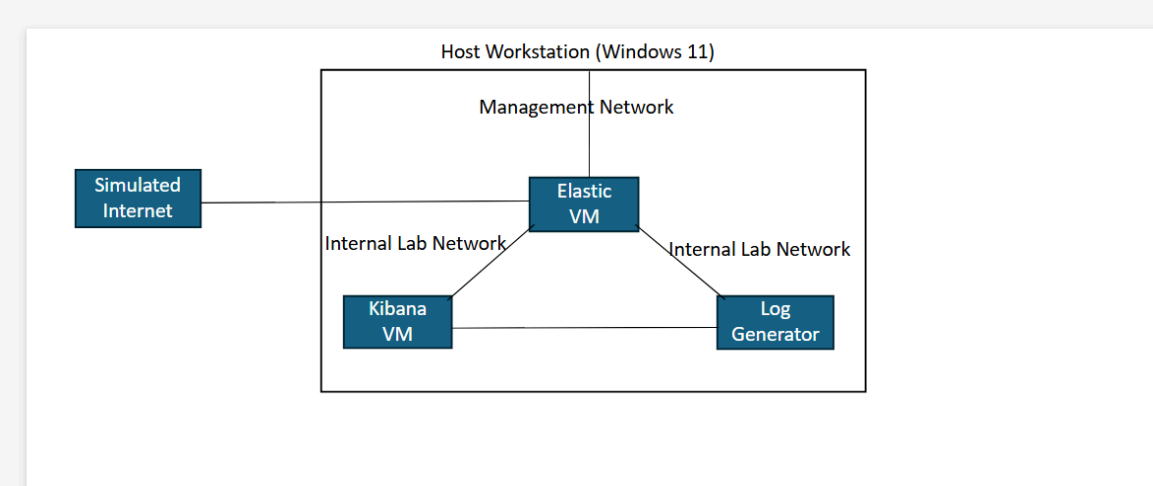
Component	Function	Host Type	OS / Version	Network Role
Elastic VM	SIEM / Data Store	Virtual Machine (Oracle VirtualBox)	Ubuntu 22.04	Internal
Log Generator	Simulated operational and threat intelligence log generation	Docker Container	Ubuntu 22.04	Internal

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Component	Function	Host Type	OS / Version	Network Role
Kibana VM	Visualization for CTI analysts	Virtual Machine (Oracle VirtualBox)	Ubuntu 22.04	Management network
Host Workstation	Overall management platform	Physical Host	Windows 11	Management network

(Replace sample data with your own.)

Attach a **diagram(s)** (Visio, draw.io, or screenshot) showing relationships between components (topology, dataflows, etc.).



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4. Hardware and Virtual Resources

List the **hardware and compute resources** supporting your testbed.

Resource	Description / Role	Specs	Allocation
Host Server	Testbed hypervisor and management platform	Intel i7-11700F (8 cores), 64 GB RAM, 3 TB SSD storage	Runs all VMs and Docker containers
VM1	Elasticsearch and Kibana	8 GB RAM, 4 vCPU, 200 GB disk	Elasticsearch service + Kibana (Visualization)

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Resource	Description / Role	Specs	Allocation
Docker Log generation container	Synthetic log and threat intelligence generation	Host resources	Simulated operational and threat intelligence log generation

(Replace sample data with your own.)

Include:

- CPU, RAM, and disk size per node
- Host operating system
- Virtualization platform (e.g., ESXi, VirtualBox)

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5. Software Stack and Tools

Provide details on all software tools installed.

Component	Version	Purpose	Installation Method
Elasticsearch	v8.13.2	Log analysis	Manual installation
Filebeat	v8.13.2	Log shipping to Elasticsearch	Manual
Python	v3.12	Create scripts to perform log generation and automated testing	Built in
Kibana	v8.13.2	Visualization for CTI analyst workflows	Manual installation
Docker container	v29.3	Log generation and simulation	Docker desktop
Oracle virtualbox	v7.2.6	Platform for testbed environment	Manual

Include any **scripts, frameworks, or libraries** (e.g., TensorFlow, Scikit-learn).

The primary analysis will be conducted using Elasticsearch and Kibana dashboards to evaluate threat prioritization and analyst efficiency. Prioritization outputs from the

contextual scoring model will be used to assess the prioritization results. Manual comparisons will evaluate analysis response times before and after automation, plus charts will be created to visualize performance improvements. Python scripts will be used for log generation and automated testing, while Elasticsearch, Kibana, and MISP-based scoring outputs (simulated) will be used for visualization (Kibana) and primary analysis.

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6. Network Configuration

Document **network setup and addressing scheme**.

Network Name	Subnet	VLAN / Segment	Purpose	Notes
Management	10.0.0.0/24	VLAN 10	Admin traffic	SSH access
Internal Lab	192.168.0.0/24	VLAN 20	Testbed data flow	Isolated
External (Simulated Internet)	172.16.0.0/24	VLAN 30	Threat simulation	NAT-only

Include:

- **Firewall rules** or **routing tables** (if applicable)- Firewall rules are configured on the Elastic VM using ufw to allow incoming traffic on port 9200, only from devices on the Internal Lab. Kibana is configured only to allow access from the Management network.
- **NAT**, **bridging**, or **port forwarding** configurations- The External (Simulated Internet) network is configured as a VirtualBox NAT network, allowing virtual machines to receive software updates while preventing outside connections from external systems. Port forwarding is disabled to prevent unprotected exposure.
- How isolation or segmentation prevents unintended exposure- The Internal lab network is configured as a Virtual Internal Network, creating a fully isolated software-defined network with no route to the host system or internet. This ensures all experimental data remains within a controlled and isolated environment.

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7. Security Controls

Summarize **security measures** implemented.

Include:

- Account and access control setup
- Use of encryption (TLS, SSH, VPN)
- Network isolation or sandboxing
- Logging/auditing measures
- Vulnerability or compliance scans (if applicable)

Example: All VMs run with non-root users; SSH access restricted to specific keys. Testbed is isolated from enterprise network via host-only adapters.

Security measures were implemented to make sure the testbed environment was isolated and ethical. User access is non-admin or non-root accounts across all virtual machines to ensure controlled privileges. SSH is used to encrypt communication between devices, reducing the risk of unauthorized access. The virtual environment is completely isolated from external networks and uses only VirtualBox network configurations, so it does not expose the host system. Logging is handled by Filebeat, which collects system and operational logs and forwards them to Elasticsearch for monitoring, auditing, and analysis. Basic system updates and scans are used to ensure all systems are up to date.

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8. Test Data Integration

Describe how data sources are integrated into the testbed.

Include:

- Data ingestion tools (e.g., Filebeat, Kafka, Logstash)
- Storage paths or indices (e.g., /var/log/testdata.log, Elastic index)
- How synthetic or real datasets are loaded
- Validation steps confirming data flow

Example: Synthetic alert data from Python scripts is streamed into Elastic via Filebeat for visualization in Kibana dashboards.

Synthetic operational and threat intelligence data is integrated into the environmental testbed using Filebeat and Python-based log generation scripts. Filebeat consumes the simulated log files and forwards them to Elasticsearch for analysis. Docker containers generate synthetic logs that mimic CTI analysts' alerts and operational data overload. Data is stored in Elasticsearch to test datasets for analysis and visualization in Kibana. Validation steps for confirming data flow include verifying log consumption in Elasticsearch and visualizing it in Kibana.

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9. Configuration Management

Document how configurations are stored and tracked.

Include:

- Configuration tools used (e.g., Ansible, Terraform, Git)
- Repository path or Git commit hash
- Backup strategy for configurations and scripts

Config Area	Tool	Location	Versioning Method
VM Templates	VirtualBox	Local Host	Snapshot
Scripts	GitLab	Local project folder	Git commits
Elastic Configs	Local config files	Ubuntu Virtual Machine	Important configuration files are manually modified and saved within the project's GitLab repository for version tracking.

10. Experiment Automation (if applicable)

Describe how experiments are executed or repeated automatically.

Include:

- Any scheduling tools (cron, Airflow, Jenkins)
- Parameters or triggers used
- Output data path and logging method

Example: Cron jobs execute data replay every 15 minutes. Each run logs system metrics and alert counts to a JSON file.

To ensure consistency across experiments, a Python master script will be used for each experimental run. The script will be manually activated at the start of each test session to ensure simulated conditions.

When executed, the script starts the Docker API synthetic log generation container, which will run for a 10-minute window. After the window closes, the script will use the Elasticsearch queries and scoring outputs to collect alert-prioritization outputs and contextualized threat scores.

Experimental metadata, such as start and end times, prioritization results, and alert counts, will be saved to a structured output file for experiment tracking. This process will ensure that each experimental run is executed and saved in the correct order for later analysis.

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11. Validation and Testing

Provide evidence that the testbed functions as intended.

Include:

- Tests run to verify data flow or alert generation
- Screenshots of dashboards or log outputs
- Example command outputs
- Observations and test results

Example: Verified data pipeline end-to-end: Log source → Filebeat → Elastic → Kibana visualization. Confirmed expected alert volumes.

The testbed was validated to confirm that data generation, visualization, and consumption all functioned correctly. Validation evidence will include screenshots of Kibana dashboards displaying log data, terminal output from log-generation scripts, and example API queries from Elasticsearch queries that showcase prioritization results and data. Synthetic log data generated using Python scripts and Docker container log generation are forwarded to Elasticsearch using Filebeat. Kibana dashboards were used to visualize log generation data in real time. Test queries are executed in Elasticsearch to verify that contextual threat data can be stored and retrieved for analysis. Testing confirmed that the environment could simulate CTI analyst workflows and the alert prioritization process.

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12. Limitations and Known Issues

Identify any **constraints** or **bugs** encountered.

Include:

- Performance limits (CPU bottlenecks, storage caps)

- Missing dependencies
- Incomplete features or manual workarounds

Example: Replay script occasionally dropped packets under high volume due to CPU contention.

The testbed is constrained by hardware limitations: it's a personal workstation, which limits the number of virtual machines and the amount of simulated log data that can be processed. A key limitation of this testbed is that core components are hosted on a limited-resource virtual environment. In a production environment, these would be deployed on separate servers to improve performance and security isolation. Datasets are synthetic and may not fully represent real-world dataset environments. Manual configuration was required for some of the tools, which may have led to inconsistent findings. Performance may be degraded by log generation rates and by CPU and memory limitations. The network is simulated using VirtualBox networking rather than higher-end networks.

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13. Backup and Recovery Plan

Describe how the environment is backed up and restored.

Include:

- Backup frequency and tools (snapshot, rsync, Git commit)
- Recovery procedures
- Test of restore process
- The testbed environment is backed up via VirtualBox VM and local files. Configuration files and scripts are stored in a Git repository to save and enable quick recall. If a system failure occurs, virtual machines can be restored from configuration files in a Git repository and from local system files. Manual recovery would include reinstalling Elasticsearch and Filebeat, also reloading the synthetic datasets to reestablish the testing environment.

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14. Change Log

Ver.	Date	Description of Change	Author
v1.0	2025-10-07	Initial setup and documentation	

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Ver.	Date	Description of Change	Author
v1.1	2025-10-10	Added automation pipeline	

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15. Mentor Review / Approval

Reviewer	Role	Date	Notes
Faculty Advisor			
DoD/ACE Mentor			
